

Erdős Problem #80

EP-80

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $c > 0$ and let $f_c(n)$ be the maximal m such that every graph G with n vertices and at least cn^2 edges, where each edge is contained in at least one triangle, must contain a book of size m , that is, an edge shared by at least m different triangles.

Estimate $f_c(n)$. In particular, is it true that $f_c(n) > n^\epsilon$ for some $\epsilon > 0$? Or $f_c(n) \gg \log n$?